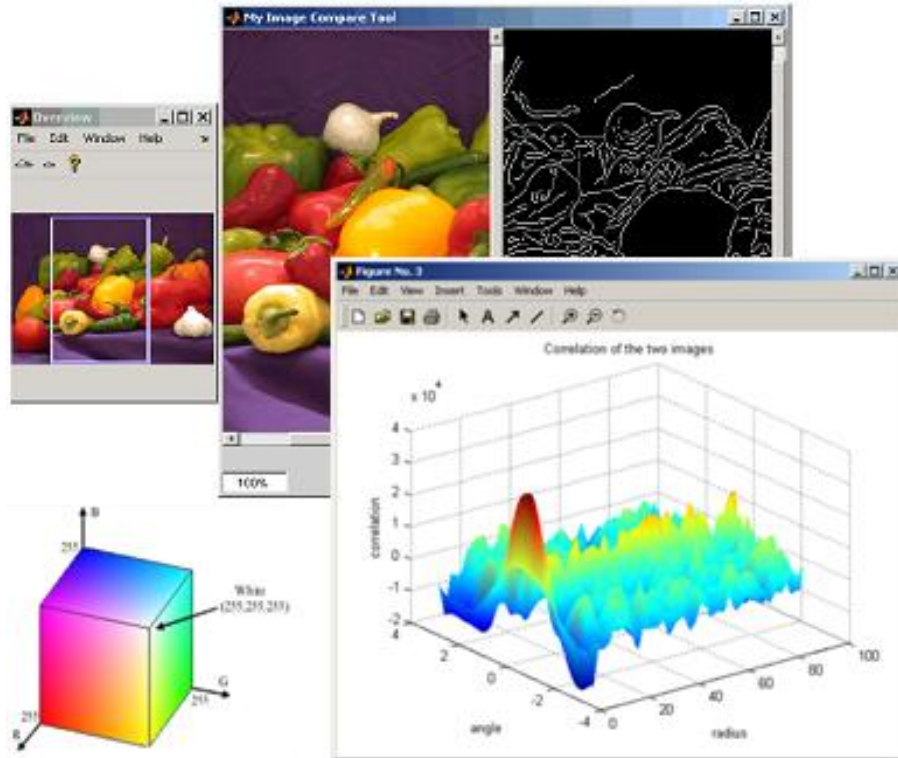


Digital Image Processing

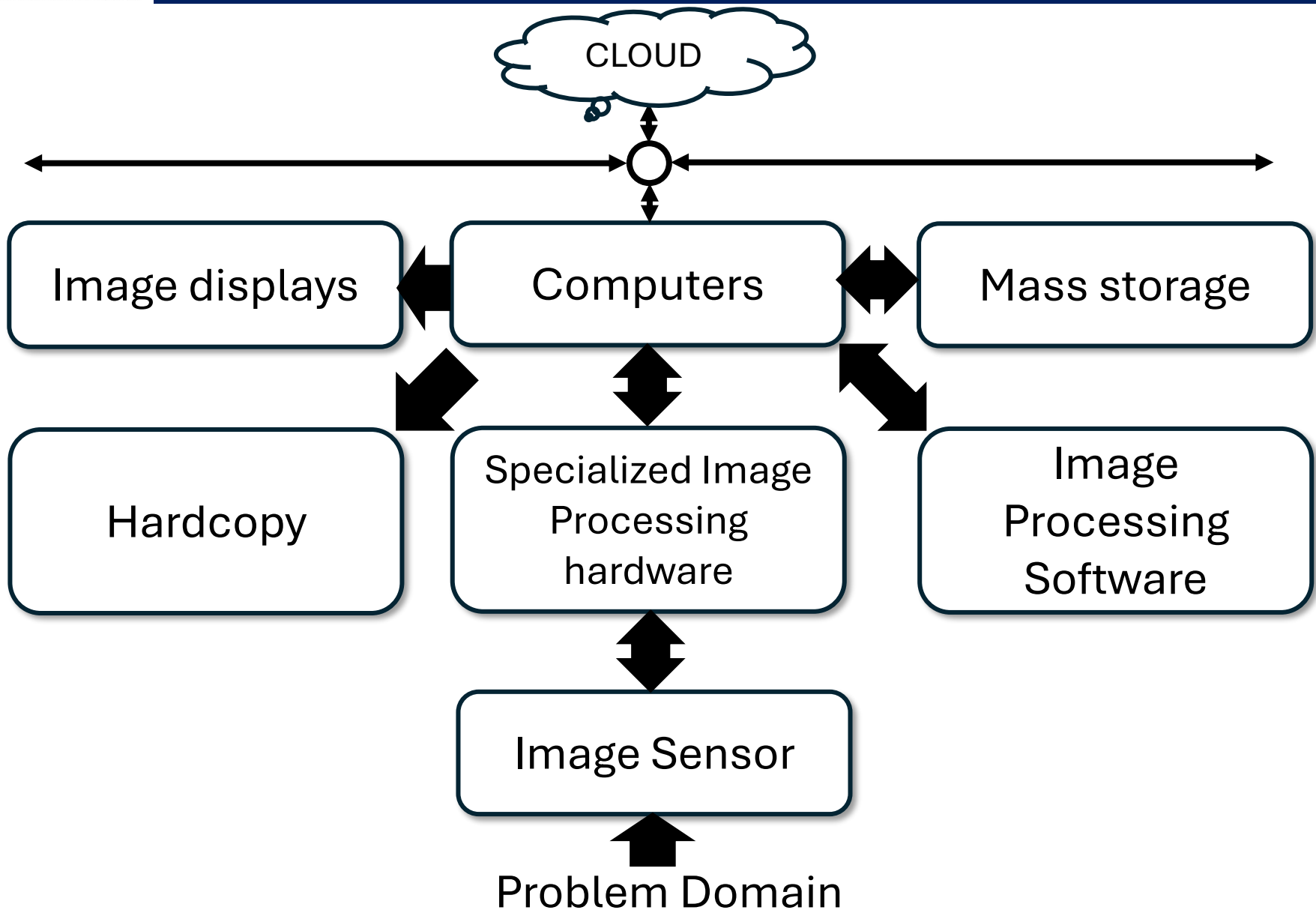
LECTURE -5: DIGITAL IMAGE COMPONENTS, BAUD RATE AND DIGITIZATION



Dr. Ajay Kumar Mahato
Assistant Professor
ECE Department
GLA University Mathura

Components of an Image Processing System

Components of Image Processing System



Components of Image Processing System

❑ Image Sensor:

- Two elements are required to acquire digital images.
 - **Physical Device** that is sensitive to the energy radiated by the object we wish to image.
 - **Digitizer**, is a device for converting the output of the physical sensing device into digital form.
- There are three main types of camera:
 - Vidicons
 - CCD (Charge Coupled Devices) camera
 - CMOS (Complementary Metal Oxide Semiconductor) camera

Components of Image Processing System

CCD Image Sensor	CMOS Image Sensor
▪ CCDs use a special manufacturing process to create the ability to transport charge across the chip without distortions.	▪ CMOS Chips use traditional manufacturing process to create the chip-the same processes used to make most microprocessors.
▪ This process leads to very high-quality sensors in terms of fidelity and light sensitivity.	▪ CMOS Sensors, are more susceptible to noise.
▪ The light sensitivity of a CCD sensors is higher.	▪ The light sensitivity of CMOS chip is tends to be lower.
▪ CCDs use a process that consumes lots of power.	▪ CMOS traditionally consumes little power.
▪ Expensive than CMOS sensors.	▪ Extermemly inexpensive compared to CCD sensors.

Components of Image Processing System



Components of Image Processing System

❑ **Specialized Image processing Hardware:**

It usually consists of digitizer just mentioned, plus hardware that performs other primitive operations, such as Arithmetic Logic Unit (ALU), that performs arithmetic and logical operations in parallel on entire image.

❑ **Computer:**

- The computer in an image processing is general purpose computer can range from a PC to a Supercomputer.
- In dedicated applications, sometime custom computers are used to achieve a required level of performance.
- However, our interest here is on general-purpose image processing system.
- In these systems, almost any well-equipped PC-type machine is suitable for off-line image processing tasks.

Components of Image Processing System

❑ Software:

- Software for image processing consists of specialised modules that perform specific tasks.
- A well –designed package also includes the capability for the user to write code that, as minimum utilizes the specialised modules.
- More sophisticated software packages allows the integration of those modules and general-purpose software commands from atleast one computer language.

❑ Mass Storage capability:

- Digital storage for image processing applications falls into three principal categories:
 - Short-term storage for use during processing.
 - On-line storage for relatively fast recall

Components of Image Processing System

❑ Image Display:

- Image display in use today are mainly color TV monitor. Monitors are driven by outputs of images and graphics display card that are an integral part of the computer system.
- In some cases, it is necessary to have stereo display, and these are implemented in the form of head-gear containing two small displays embedded in googles worn by user.

❑ Hardcopy Devices:

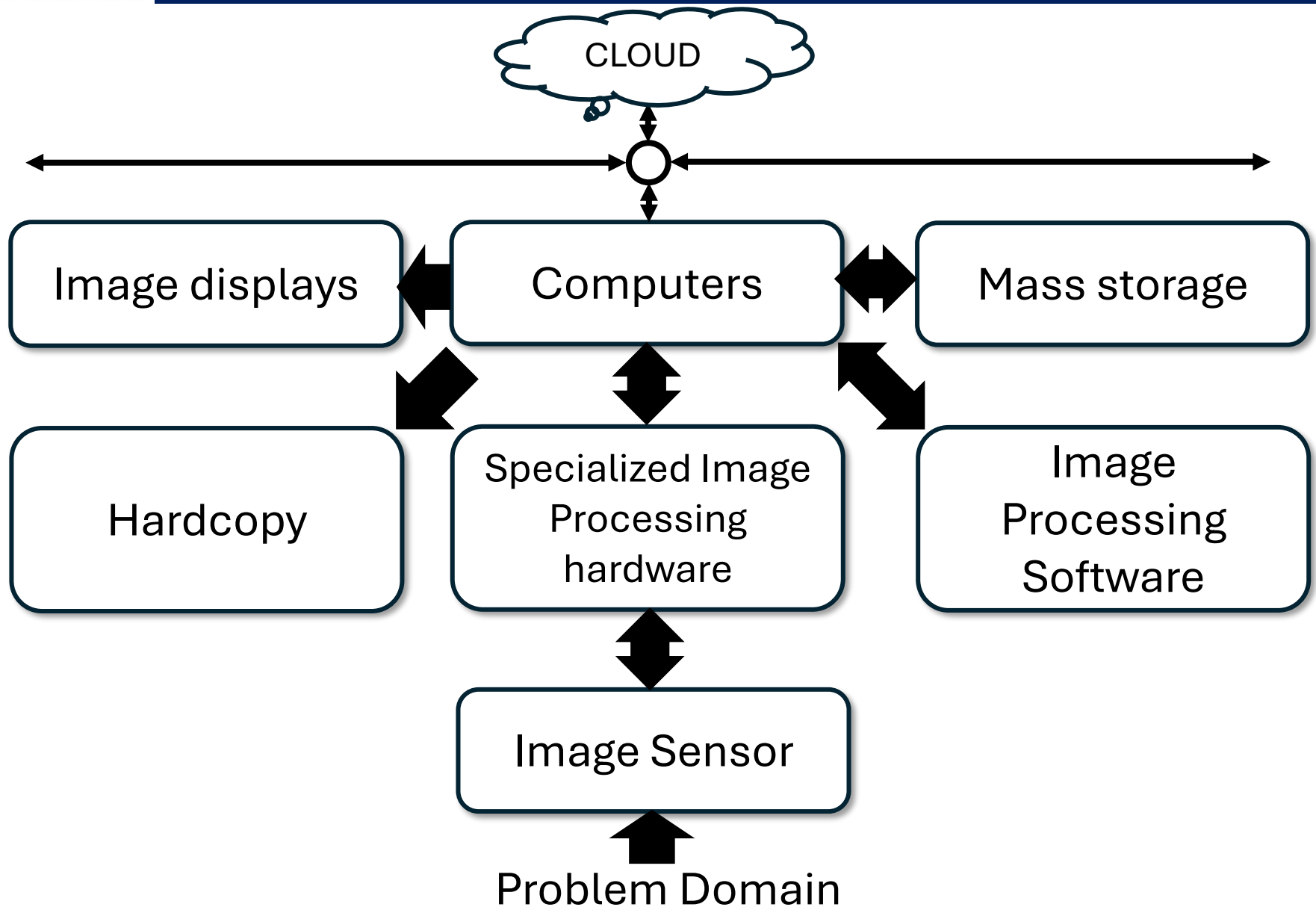
- Hardcopy devices for recording images include laser printer, film cameras, heat-sensitive devices, inkjet units, and digital units such as optical and CD-ROM disks.
- Film provides the highest possible resolution, but paper is the obvious medium of choice for written material.

Components of Image Processing System

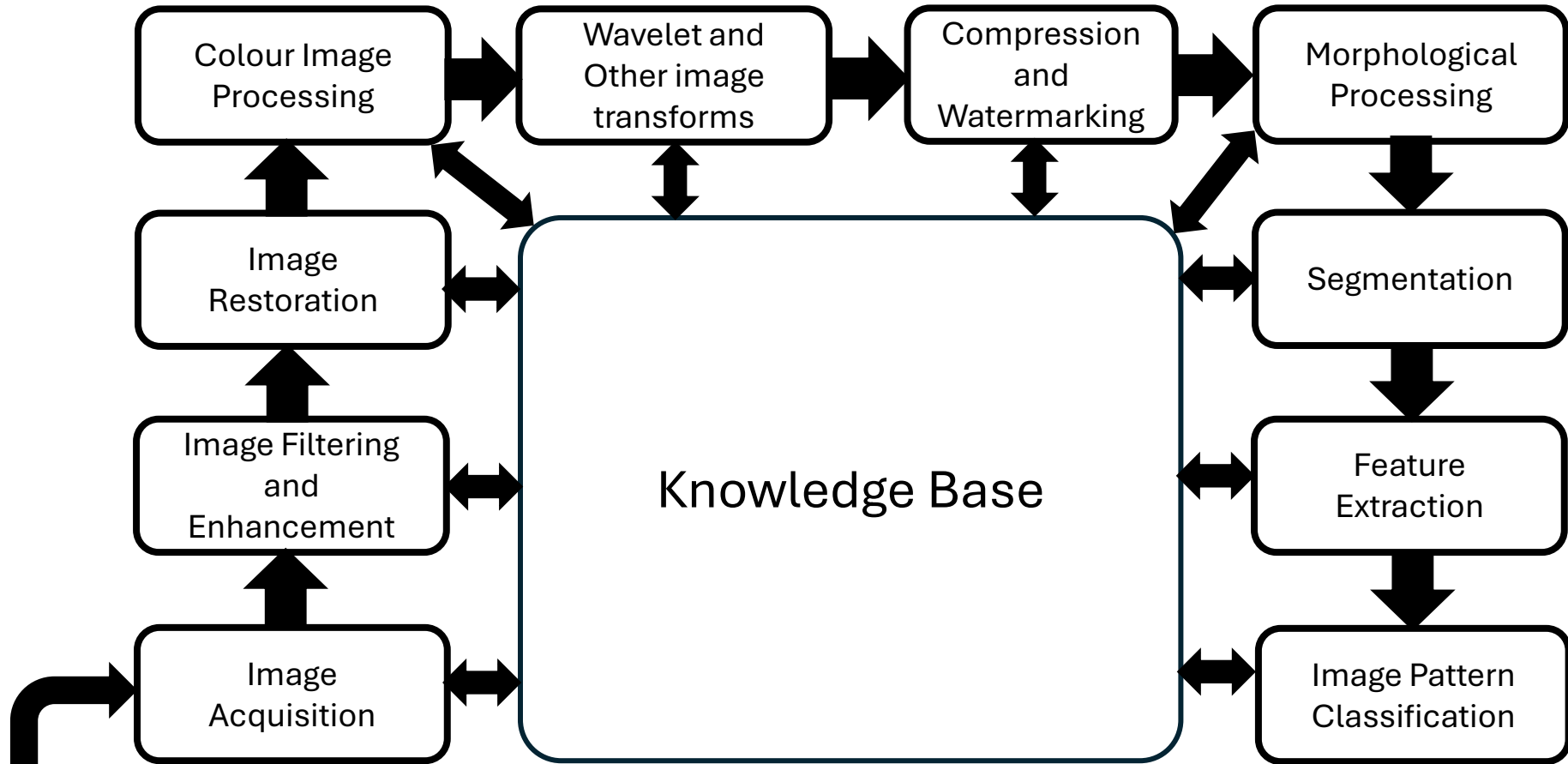
❑ Networking:

- Networking is almost a default function in any computer system in use today.
- Because of large amount of data inherent in image processing applications, the key consideration in image transmission is bandwidth.
- In dedicated networks, this typically is not a problem, but communication with remote sites via the internet are not always as efficient.
- Fortunately, this situation is improving quickly as a result of optical fibre and other broadband technologies.

Components of Image Processing System



Fundamental Steps in Image Processing



Problem Domain

University Questions

Ques 1: Explains the steps in image processing with the help of block diagram.

GLA University, Mid Term Examination, Odd Semester 2024-25

Ques 2: What is Digital Image Processing? Explains fundamental steps in Digital Image Processing.

GLA University, End Term Examination, Even Semester, 2021-22

Ques 3: Explain the steps in image processing with the help of block diagram.

GLA University, Summer term Carry-Over Examination, Session 2023-24



Image Baud Rate

Image Baud Rate

- **Baud Rate:** Baud rate, on the other hand, denotes the **number of symbols or signal changes transmitted per second** on the channel
- **Bit Rate:** Bit rate refers to the **number of bits transmitted per second** in a communication channel, which represents the actual data transfer rate
 - R: represents the number of bits per sec.
 - r: represents the number of symbols per sec or number of elements per sec
 - n: represents the number of bits per symbol or number of bits per element
- Mathematically, we may write:

$$\text{Bit Rate} = \text{Baud Rate} \times \text{Bits per Symbol}$$

$$R = r \times n$$

Image Baud Rate

Ques1: Consider the followings:

- Grayscale image consists of 256 pixels, and each pixel is represented using 8 bits.
- If the transmission occurs at a baud rate of 2400 baud, and assuming each symbol equals one bit

Determine the time required to transmit the image completely.

Image Baud Rate

Ques2: let's use a modulation scheme where each symbol encodes 4 bits. With a baud rate of 2400 baud.

Determine the (A) bit rate and (B) transmission time for a 2048-bit image.

Image Baud Rate

Ques3: You are transmitting a grayscale image using a serial communication link with a baud rate of 9600 baud. The image has a resolution of 640x480 pixels. Each pixel is represented by 1 byte (8 bits). Assuming each symbol represent 1 bit, Determine the (A) bitrate and (B) time required to transmit the image.

Image Baud Rate

Ques4: Let's consider a high-resolution color image with a resolution of 1920×1080 pixels. Each pixel uses 3 bytes (24 bits) for color information (RGB). The baud rate is 384,000 baud, and each symbol represents 4 bits. Determine the (A) Bit rate and (B) time required to transmit the image.

Image Digitization (Sampling and Quantization)

Image Digitization

$$f(x, y)$$

Where

x, y : represents the spatial coordinates

f: represents the amplitude/intensity

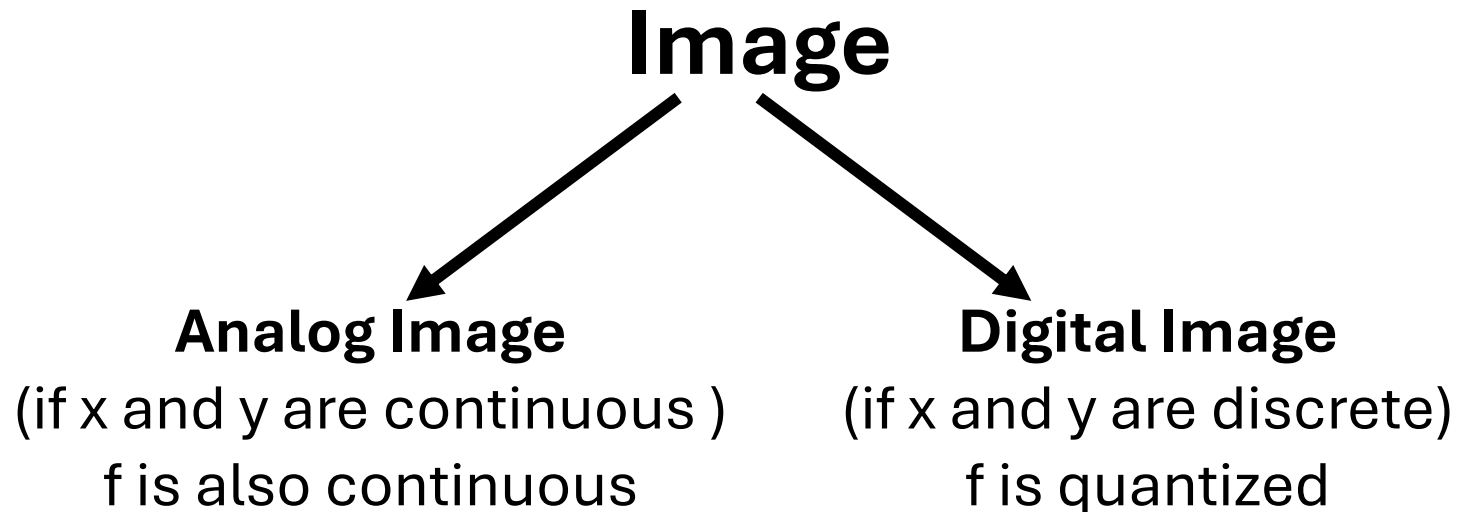
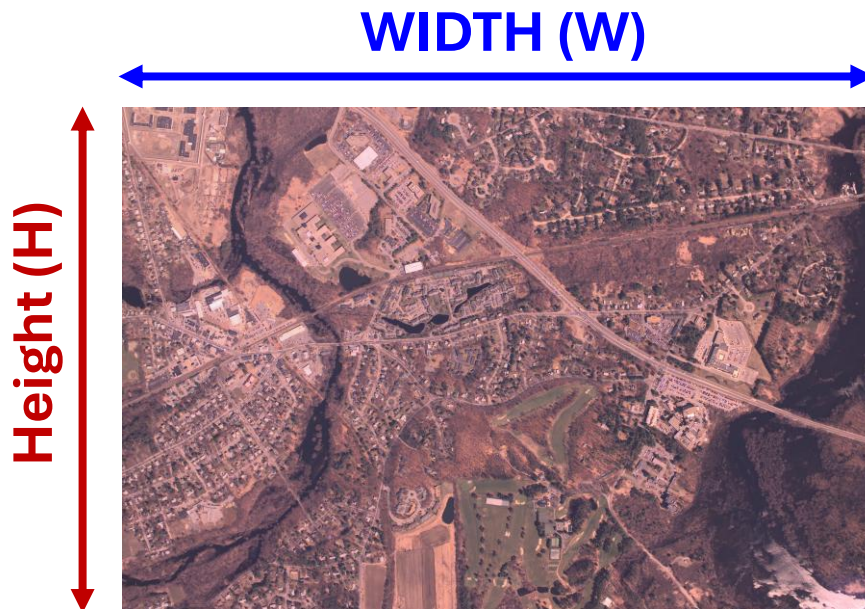


Image Digitization

❑ Challenges with Analog Images:



$$0 \leq x \leq H$$

$$0 \leq y \leq W$$

$$I_{\min} \leq f(x,y) \leq I_{\max}$$

Image Digitization

❑ Theory of Real Numbers:

- Between any two given points, there are infinite number of points
- An should be represented by infinite number of points
- Each such image points may contain one of the infinitely many possible intensity/colours values needing infinite number of bits.

Obviously, such a representation is not possible in any digital computer.

Image Digitization

□ What is desired:

- An image is to be represented in the form of a finite 2D-matrix

$$f(x, y) = \begin{bmatrix} f(0, 0) & f(0, 1) & f(0, 2) & \dots & \dots & f(0, N - 1) \\ f(1, 0) & f(1, 1) & f(1, 2) & \dots & \dots & f(1, N - 1) \\ \vdots & \vdots & \vdots & \ddots & \ddots & \vdots \\ f(M - 1, 0) & f(M - 1, 1) & f(M - 1, 2) & \dots & \dots & f(M - 1, N - 1) \end{bmatrix}$$

Each of the matrix elements should assume one of the finite discrete values

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Image Digitization

- **Step1:** Digitizing the coordinates i.e. spatial locations called as Sampling.
- **Step 2:** Digitizing the amplitude i.e. intensity levels is called as Quantization.

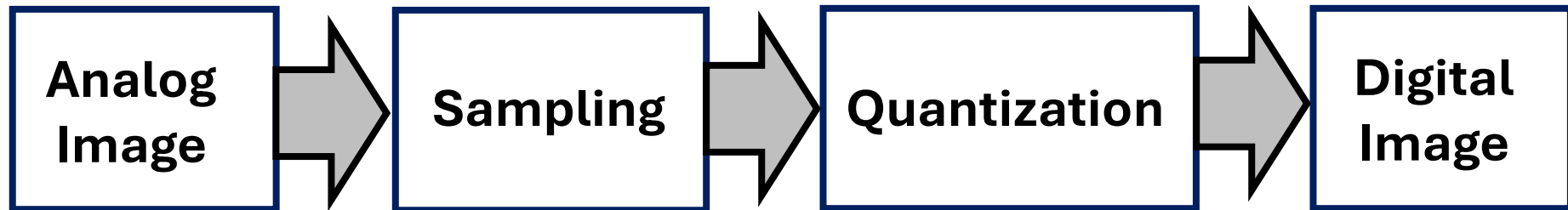
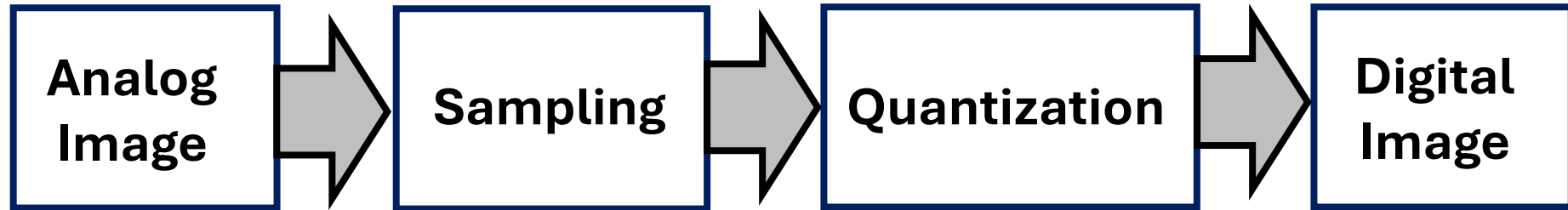


Image Digitization



❖ Image digitalization consists of two steps:

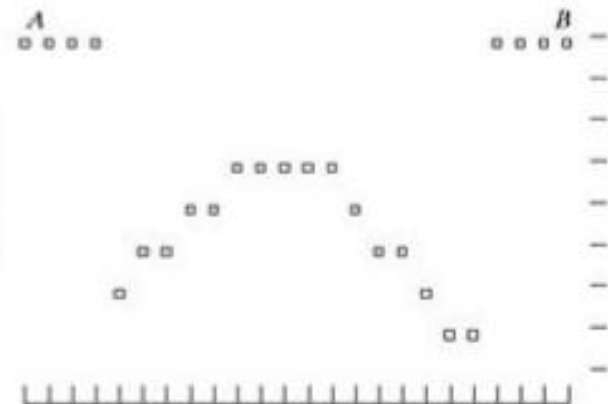
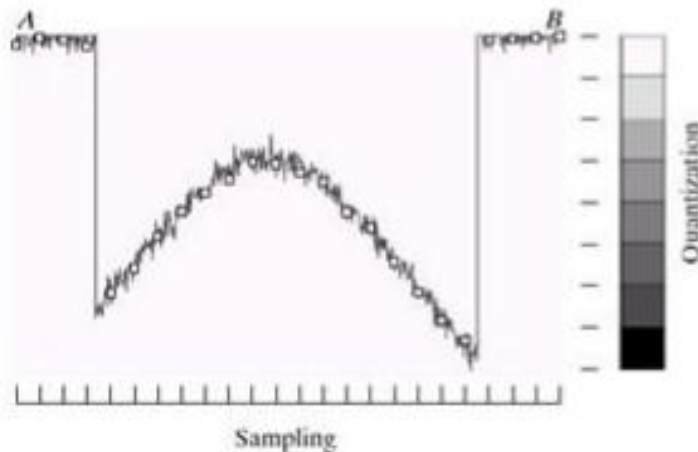
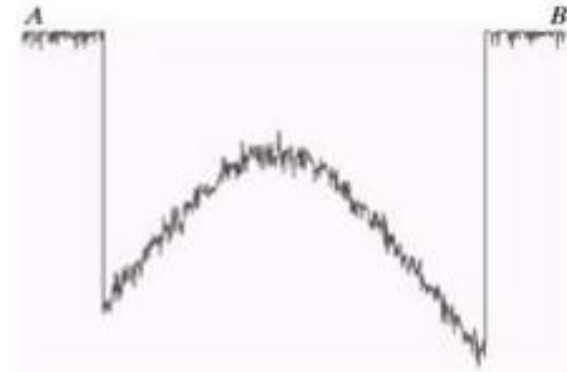
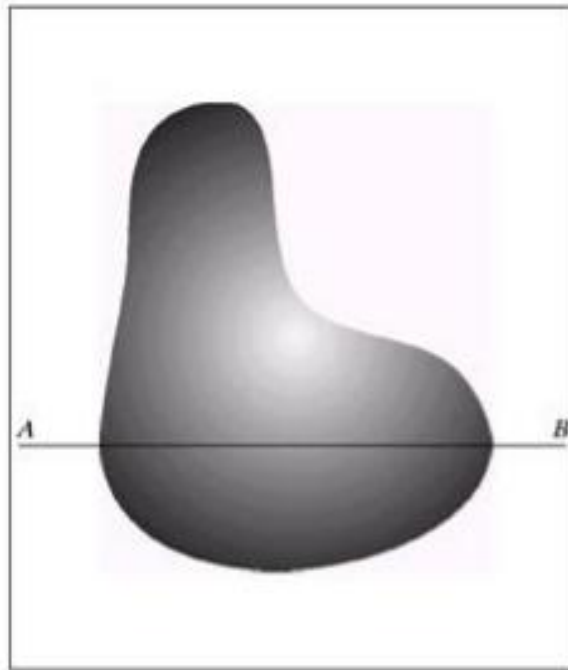
❑ Sampling:

- Digitizing the coordinates i.e. spatial locations called as Sampling.
- Image representation by 2D finite matrix

❑ Quantization:

- Digitizing the amplitude i.e. intensity levels is called as Quantization.
- Each matrix element represented by one of the finite set of discrete values

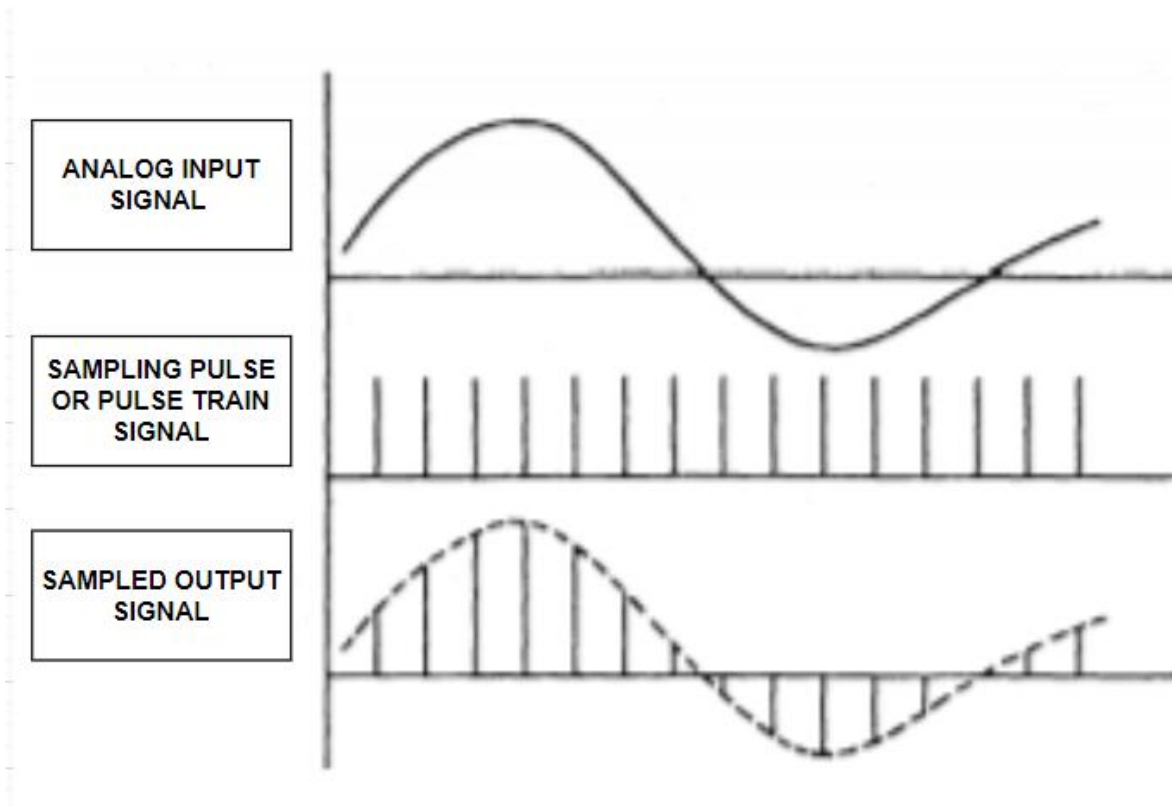
Image Digitalization



Sampling

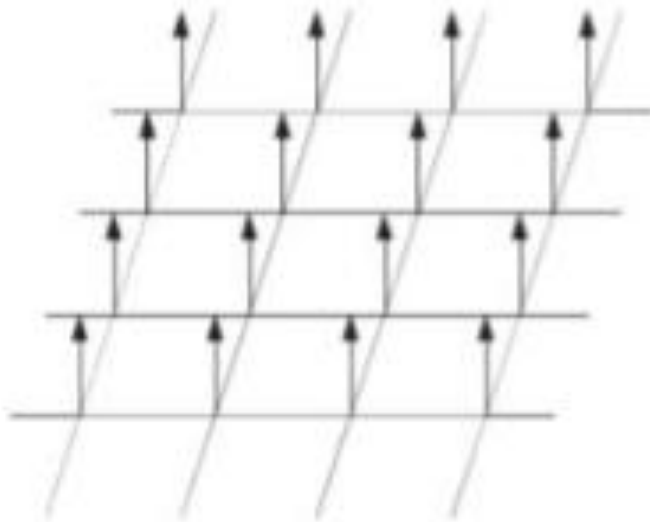
- Sampling process converts an analog signal into a discrete signal.
- Image representation by 2D finite matrix
- Digitizing the coordinate values is called sampling.

❑ One-Dimensional Sampling:



Sampling

❑ Two-Dimensional Sampling:



A comb of delta functions

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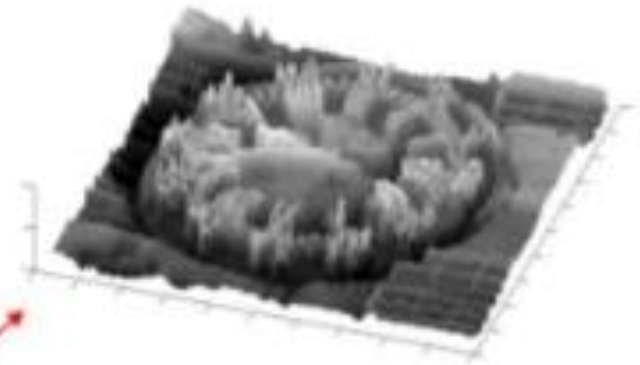


Image as a 2D function / surface



Quantization

- Digitizing the amplitude values is called quantization.
- It is the process of converting sampled analog pixel intensity into a discrete variable integer.
- Each matrix element represented by one of the finite set of discrete values

Quantization is a mapping of a continuous variable u to a discrete variable U'

$$U' \in \{r_1, r_2, r_3 \dots \dots \dots r_L\}$$



Quantization

Mapping is generally a staircase function

Quantization Rules:

Define a set of decision or transition levels $(t_k, k= 1, 2, ..L+1\}$,

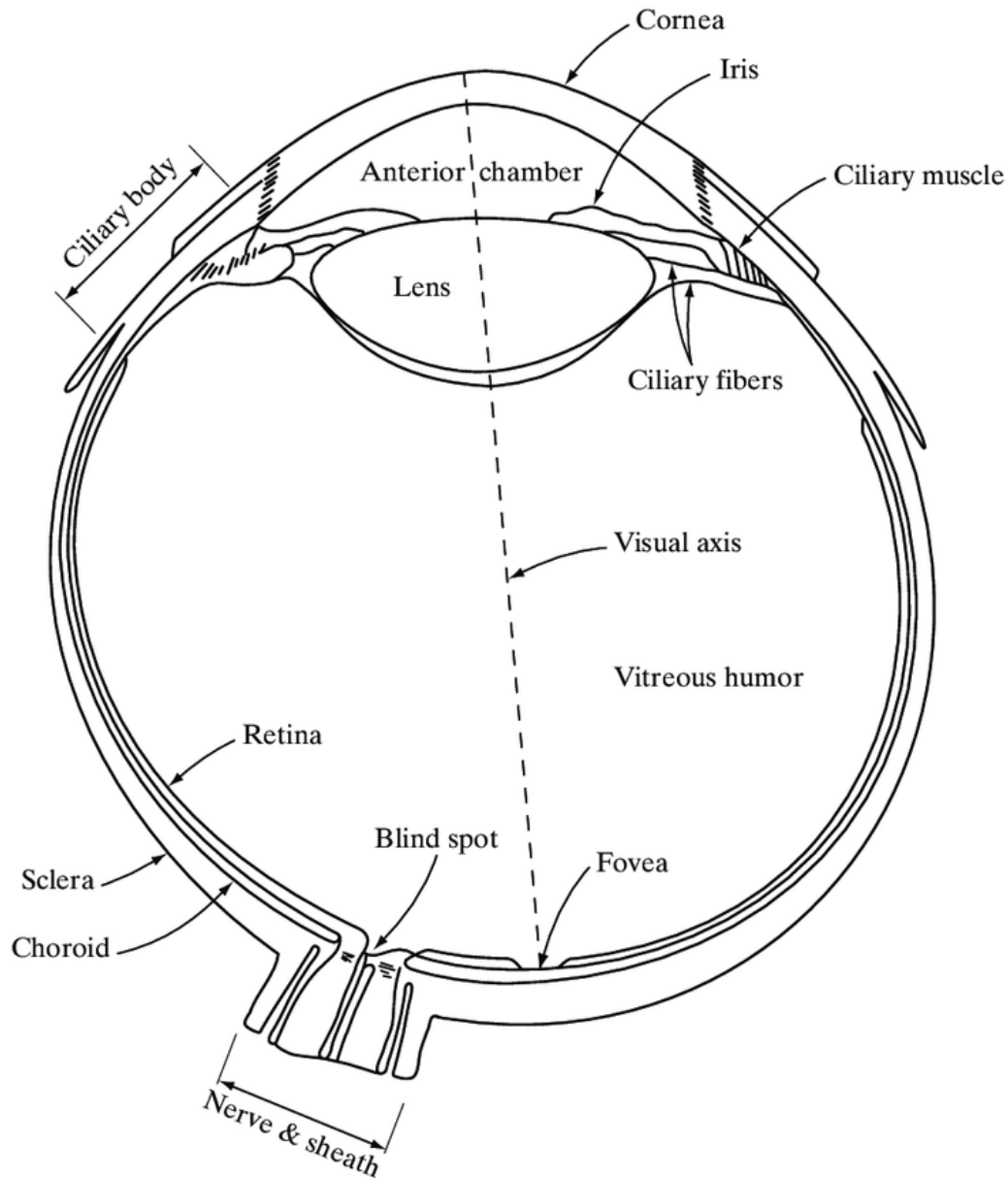
$$U' = r_k \quad \text{if } t_k < u < t_{k+1}$$

Where

- t_1 : is the minimum value
- t_{L+1} : is the maximum value

Elements of Visual Perception

Structure of Human Eye



Structure of Human Eye

- The eye is nearly a sphere (with a diameter of about 20 mm) enclosed by three membrane:
 - The CORNEA and the SCLERA outer cover
 - The CHOROID
 - The RETINA
- **CORNEA:** The CORNEA is tough, transparent tissue that covers the anterior surface of the eye.
- **SCLERA:** Continuous with the CORNEA, the SCLERA is an opaque membrane that encloses the remainder of the optic globe.
- **CHOROID:** The CHOROID directly below the SCLERA. This membrane contains a network of blood vessel that serve as the major source of nutrition to the eye.

Structure of Human Eye

■ LENS:

- The LENS consists of concentric layers of fibrous cells and is suspended by fibres that attach to the ciliary body.
- In the extreme cases, excessive clouding of the LENS, referred to as cataracts, can lead to poor colour discrimination and loss of clear vision.
- The LENS absorbs approximately 8% of the visible light spectrum, with higher absorption at shorter wavelength.
- Both infrared and ultraviolet lights are absorbed by proteins within the lens and in excessive amounts can damage the eye.

Structure of Human Eye

❑ RETINA:

- The innermost membrane of the eye is the RETINA, which lines the inside of the wall's entire posterior portion.
- When the eye is focussed, light from an object is imaged on the retina.
- Pattern vision is afforded by discrete light receptors distributed over the surface of the retina. There are two types of receptors:
 - CONES
 - RODS

Structure of Human Eye

❑ CONES:

- There are between 6 and 7 million cones in each eye. They are located in the central portion of the retina called the fovea and are highly sensitive to colour.
- Humans can resolve fine details because each cone is connected to its own nerve end. Cone vision is called photopic or bright-light vision.

❑ RODS:

- There are between 75 to 150 million rods are distributed over the retina. Rods capture an overall image of the field of view.
- They are not involved in colour vision and are sensitive to low level of illumination.

Structure of Human Eye

FIGURE 2.2
Distribution of rods and cones in the retina.

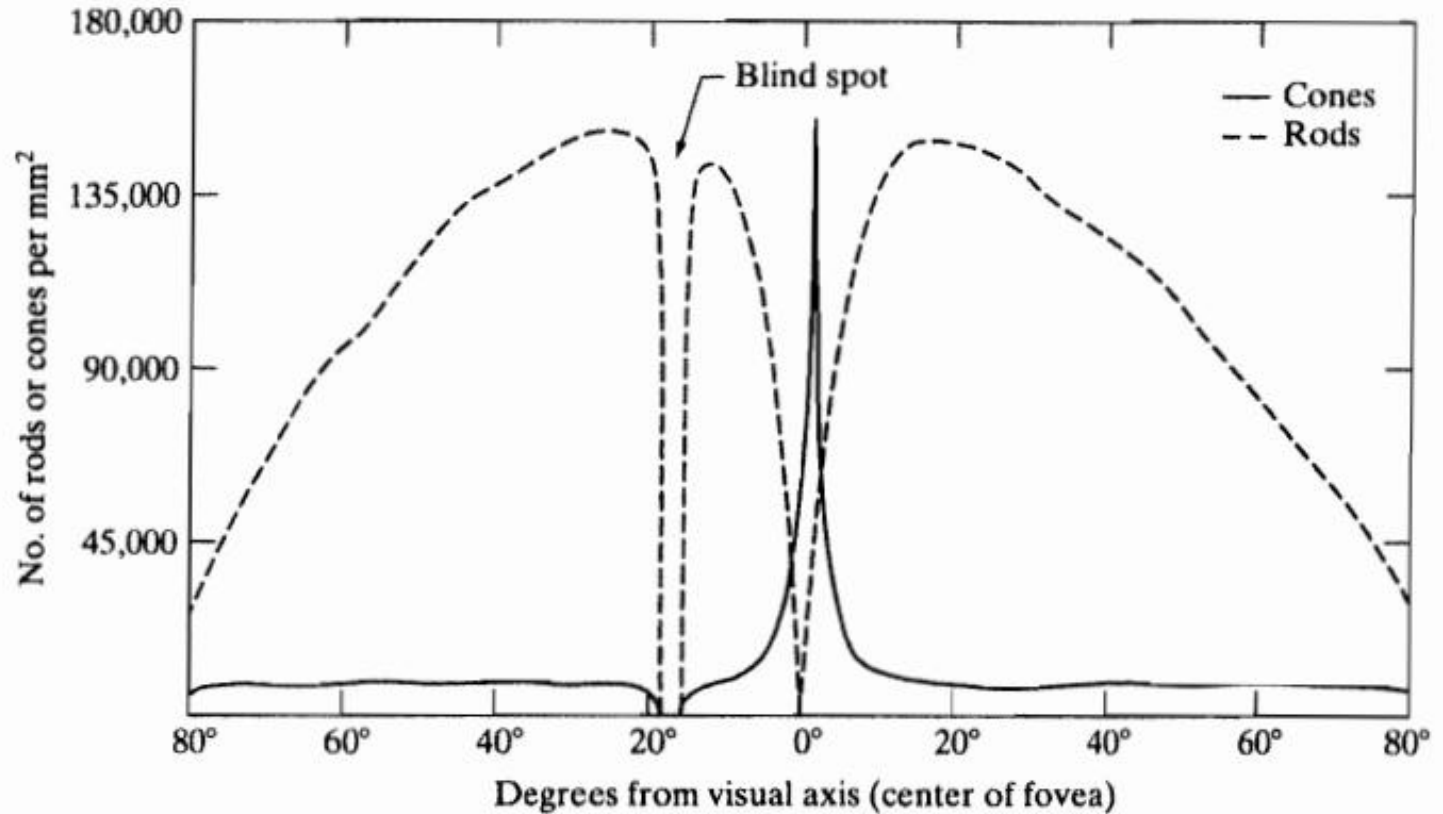


Figure shows the density of rods and cones for a cross section of the right eye, passing through the region where the optic nerve emerges from the eye. The absence of receptors in this area causes the so-called **blind spot**.

Image Formation in Eye

- In an ordinary photographic camera, the lens has a fixed focal length. Focussing on various distances is achieved by varying the distance between the lens and the imaging plane, where the film is located.
- In the human eye, the converse is true; the distance between the center of the lens and imaging sensor (retina) is fixed, and the focal length needed to achieve proper focus is obtained by varying the shape of the lens.
- The fibers in the ciliary body accomplish this by flattening or thickening the lens for distant or near objects respectively.
- The distance between the center of the lens and the retina along the visual axis is approximately 17 mm. The range of the focal length is approximately 14 mm to 17 mm, the latter taking place when the eye is relaxed and focussed on distances greater than about 3m.

Image Formation in Eye

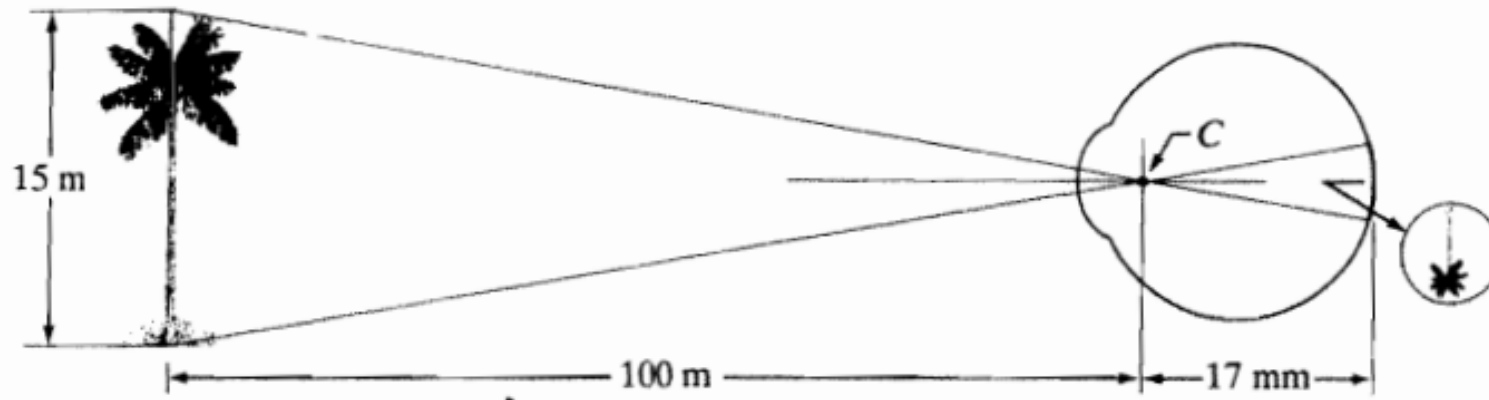


FIGURE 2.3
Graphical representation of the eye looking at a palm tree. Point C is the optical center of the lens.

$$\frac{15}{100} = \frac{h}{17} \quad h = \frac{15 * 17}{100} = 2.5 \text{ mm}$$

As indicated earlier in this section, the retinal image is focussed primarily on the region of the fovea. Perception then takes place by the relative excitation of light receptors, which transform radiant energy into electrical impulses that ultimately are decoded by the brain



Thank You